



INTRODUCTION TO

# engineering & THE environment

**EDWARD S. RUBIN**

with **CLIFF I. DAVIDSON**

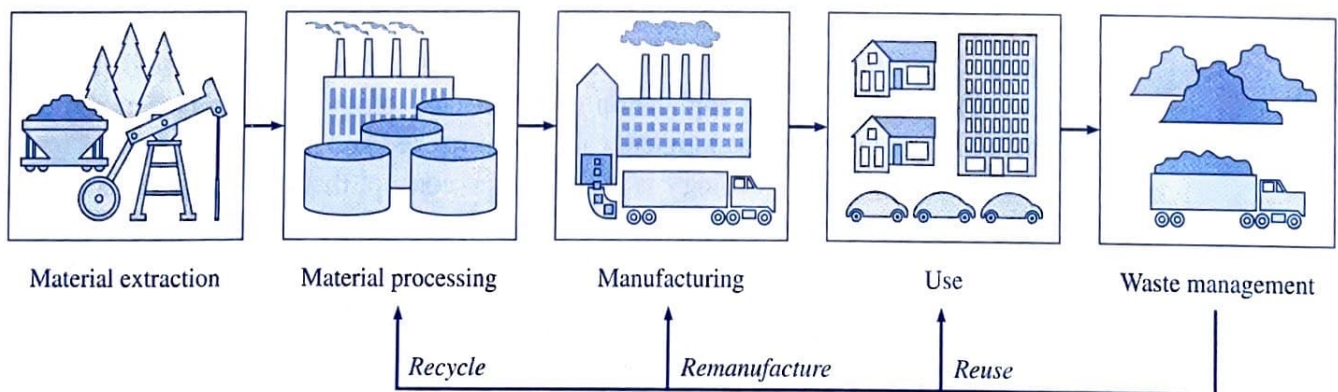
**Energy Use** This source of environmental impacts is perhaps the most pervasive and most important of any that engineers deal with. Energy is vital for life and for an economy, and the quantities and types of energy that a society uses directly affect environmental quality.

Energy use encompasses everything from the heating and cooling of homes and buildings, to the electricity that runs modern computers and appliances, to the gasoline and other fuels that power our transportation system. The manufacturing processes discussed earlier all require energy to perform their tasks. Most of the world's energy today comes from fossil fuels—oil, coal, and natural gas—and when these fuels are burned or converted to electricity the environment is affected, as you will see in Chapter 2. Nuclear power leaves a different kind of environmental legacy, and even renewable energy sources like hydroelectric, biomass, solar energy, and wind power are not without their adverse environmental consequences.

A good rule of thumb is that any engineering improvement that reduces the energy required for a particular service will be beneficial for the environment. Thus a desktop computer designed to use 400 watts of power instead of 800 watts, or a car that gets 30 miles per gallon (mpg) instead of 20 mpg, will be environmentally friendlier. Switching from one energy form to another—e.g., from gasoline-powered cars to electric-powered cars—is a lot trickier to deal with. As we shall see in later chapters, the environmental consequences may involve trade-offs that are difficult to evaluate.

## 1.5.2 A Life Cycle Perspective

An *environmental life cycle assessment (LCA)* provides the “big picture” of how engineering decisions in any particular area affect the environment. Figure 1.3 illustrates the connections among the stages commonly involved in building and creating goods to serve society. Environmental impacts occur at each stage as a result of materials consumption and transformation, and the use of energy.



**Figure 1.3** Stages of a product life cycle. Environmental impacts occur at all stages. These impacts can be reduced by engineering designs that change the type and/or amount of materials used in a product; by creating more efficient extraction and manufacturing processes; and by improving the recovery and reuse of materials and energy at the end of the product life. (Source: Based on OTA, 1992)

Traditionally, most engineers focus on only small pieces of this overall system. But this is starting to change. Environmental awareness has made it clear that all stages of a product's life cycle must be considered in finding ways to reduce environmental impacts. Improvements result from creating cleaner, more efficient manufacturing operations; from reducing the energy and materials needed for use of a product; and from improving the recovery of energy and materials during waste management. Life cycle assessments are thus an important tool in implementing the concepts of green design, pollution prevention, and waste minimization that are becoming fundamental paradigms of good engineering practice.

### 1.5.3 Industrial Ecology and Sustainable Development

The perspectives of a life cycle assessment provide the foundation for a more comprehensive view of design for the environment that has come to be known as *industrial ecology*. This term has gained popularity in recent years as an embodiment of the environment principles that should be reflected in engineering design. As defined by Graedel and Allenby (1995),

Industrial ecology is the means by which humanity can deliberately and rationally approach and maintain a desirable carrying capacity, given continued economic, cultural, and technological evolution. The concept requires that an industrial system be viewed not in isolation from its surrounding systems, but in concert with them. It is a systems view in which one seeks to optimize the total materials cycle from virgin material, to finished material, to component, to product, to obsolete product, and to ultimate disposal. Factors to be optimized include resources, energy, and capital.

This view is very much in keeping with the concept of *sustainable development*, which also has been widely popularized in recent years. As originally defined by the World Commission on Environment and Development (WCED, 1987),

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

This statement reflects a strong moral and ethical position about our own responsibility to future generations. But how does one operationalize this goal? Allenby (1999) suggests that the application of industrial ecology principles offers a means by which sustainable development can be approached and maintained. In more practical terms, industrial ecology is based on the concept that natural systems tend to recirculate and reuse materials, thus eliminating or minimizing the production of wastes and the use of energy. This offers a lesson and a path for the development of industrial systems and technology. Another embodiment of industrial ecology principles was offered in a recent National Research Council study (NRC, 1996) that suggests that industrial ecology should include (1) circulating and reusing material flows within the system; (2) reducing the amount of materials used in products to achieve a particular function; (3) protecting living organisms by minimizing or eliminating the flow of harmful substances; and (4) minimizing the use of energy and the flow of waste heat back to the environment.